Power Electronics Modelling and Simulation of 2025 BMW i4 eDrive35 Gran Coupe Electric Vehicle Powertrain (2025)

Glynnly Masanga, Qinghao Zeng, Luyao Zhou

Abstract*—This report presents a comprehensive simulation and design study of key power-electronic subsystems for the 2025 BMW i4 eDrive35 Gran Coupe. The project consists of three phases: vehicle energy modeling, three-phase inverter design, and on-board charger development. A longitudinal energy model first establishes a baseline for miles-per-gallon equivalency (MPGe = 174.1) and well-to-wheel CO_2 emissions, revealing strong regional variation from 3.48 to 170.7 gCO_2/mi depending on grid mix. Building on this baseline, Phase 2 designs a 210 kW three-phase voltage-source inverter using Silicon-Carbide (SiC) MOSFET technology, achieving 98.45% efficiency at full load, with loss and waveform analysis validated in Simulink under 100%, 50%, and 25% load conditions. Phase 3 develops a two-stage on-board charger comprising a boost PFC front-end and isolated DC–DC converter, implementing commercial devices (CAB650M17HM3, IPDQ65R008CM8, STTH6010) and demonstrating 92.82% overall efficiency at 7.2 kW. The results confirm strong agreement between analytical and simulated data and highlight how SiC-based inverter and charger topologies can substantially enhance BEV efficiency and carbon performance.

***Keywords*—Battery Electric Vehicle, Inverter Design, Battery Charger, Power Electronics, SiC MOSFET, Emissions Analysis, Energy Efficiency**

I. INTRODUCTION

A. Sustainable Development Goals and the Global Transition to Battery Electric Vehicles

The global transition toward electric mobility has accelerated the development of high-efficiency power electronic converters for automotive applications. With the world working together to try and achieve the Sustainable Development Goals, there is a need for more efficient technologies and less carbon-emitting vehicles [1]. However, the environmental benefits and operational performance of BEVs are heavily dependent on powertrain component design, regional electricity grid composition, and charging infrastructure characteristics.

The BMW i4 eDrive35 Gran Coupe represents a modern premium battery electric vehicle (BEV) platform featuring a 66-kWh usable capacity Li-Nickel Manganese Cobalt battery capacity, a 210 kW electric motor, and an approximately 305-mile EPA range. Understanding the complete energy conversion chain—from grid electricity through charging systems, battery storage, power electronics, and motor drive—is essential for optimizing vehicle efficiency and minimizing environmental impact.

B. Project Objectives

This project addresses three fundamental aspects of BEV powertrain and analysis:

1. Vehicle Characterization and Emissions Analysis: Establish comprehensive vehicle parameters, validate fuel economy calculations through multiple methodologies, and quantify regional emissions variations based on electricity grid carbon intensity.
2. Inverter Design and Optimization: Design a high-efficiency three-phase voltage source inverter capable of delivering rated motor power with minimal losses across varied operating conditions, utilizing advanced wide-bandgap semiconductor technology.
3. Battery Charger Development: Develop a compact AC-DC charging system with active power factor correction and galvanic isolation, utilizing a two-stage topology consisting of a Boost PFC front-end and an isolated Full-Bridge DC-DC converter. The objective is to deliver a regulated 7.3 kW, 400 V output compatible with the BMW i4 battery platform, while ensuring unity power factor and minimizing switching losses through the selection of Superjunction MOSFETs and optimized dual-loop control strategies.

C. Report Organization

The format of this report is as follows: In Section II, theoretical underpinnings are established and pertinent literature is reviewed. Each project phase's methodology and component selection is described in Section III. Results from computations and simulations, methodology comparisons, and practical considerations are covered in Section IV. Conclusions are provided at the end of Section V.

II. LITERATURE REVIEW

A. BEV Energy Consumption and Emissions

The energy consumption of BEV is governed by road-load forces opposing motion and powertrain conversion losses. The instantaneous tractive force required is expressed as

$$F_{res} = A + Bv + Cv^2$$

Where A represents constant rolling resistance and accessory loads (N), B is speed-dependent rolling resistance (Nsm^{-1}), C is aerodynamic drag coefficient Ns^2m^{-2} and v is vehicle velocity (m/s) [2]. Th^i^e quadratic relationship indicates that aerodynamic losses dominate at highway speeds, while rolling resistance is significant during urban driving. Well-to-wheel

(WTW) emissions analysis accounts for electricity generation, transmission, and distribution losses. Regional grid carbon intensity varies significantly, from approximately 20gCO$_2$/kWh in renewable energy-dominated grids (Norway) to 882 gCO_2/kWh in non-renewable or fossil fuel-intensive regions (West Virginia). Massachusetts maintains a moderate grid intensity of $354gCO_2$ through a balance between natural gas and renewable generation.

B. Three-Phase Voltage Source Inverters

Voltage source inverters (VSIs) convert DC battery voltage to variable-frequency three-phase AC used for Motor control in BEVs'. Another common inverter is a Current Source Inverter (CSI), and the main difference is that in VSIs the voltage source is stiff while in CSIs the current is stiff. However, VSIs allow us to have a smaller inductor, and that is why they are used in most BEVs. Space vector pulse width modulation (SVPWM) is widely adopted for automotive applications due to superior DC bus utilization, 15% higher voltage compared to sinusoidal pulse width modulation (Sine PWM), and a reduced harmonic current [3].

Silicon Carbide (SiC) MOSFETs offer substantial advantages over traditional silicon IGBTs for automotive inverters: near zero reverse recovery charge, lower on-state resistance, higher switching frequencies, and superior high-temperature operation. These characteristics enable higher efficiency, increased power density, and reduced cooling requirements.

Inverter losses comprise conduction losses (I^2R losses during on-state) and switching losses (energy dissipated during transitions). The total MOSFET conduction loss is:

$$P_{cond,MOS} = I^2_{MOS,RMS} * rds_{on}$$

Switching losses scale linearly with switching frequencies and are extracted from manufacturer energy loss data and are calculated as:

$$P_{sw,MOS} = (f_s/2)(E_{on} + E_{off})(V_{dc}/V_{test})$$

The SiC MOSFET switch has a body diode with a very low reverse conduction compared to traditional diodes; however, the body diode has reverse recovery losses calculated as:

$$P_{cond,D} = V_f * I_{D,avg} * t_{dead}/T_{sw}$$

$$P_{rr} = f_{sw} * E_{rr} * V_{dc}/V_{test}$$

C. Battery Charging Systems

On-board chargers (OBCs) for electric vehicles commonly adopt a two-stage architecture consisting of a power-factor-correction (PFC) front-end followed by an isolated DC-DC converter. Recent studies have shown that boost-type PFC circuits operating in continuous-conduction mode achieve high power factor and reduced input harmonic distortion for 3-7 kW residential charging applications [4]. For the isolation stage, full-bridge and LLC resonant converters are the most widely used topologies, offering high efficiency and reliable galvanic isolation for 400V and 800V battery systems [5]. Prior research also highlights the importance of semiconductor device selection, particularly the trade-offs between Si MOSFETs, SiC MOSFETs, and fast-recovery diodes, as these devices strongly influence conduction and switching losses in high-frequency OBC designs [6].

Furthermore, several works have emphasized that accurate loss modeling requires incorporating datasheet parameters such as RDS(on), Coss, and Qrr into simulation environments rather than relying on ideal switching behavior. Building on these established methods, the present project implements a boost PFC stage combined with an isolated full-bridge DC–DC converter and evaluates semiconductor performance through detailed datasheet-based Simulink modeling [7].

III. METHODOLOGY

A. Phase 1: Vehicle Characterization

1. Parameter Acquisition

The EV Database, EPA certification paperwork, and manufacturer datasheets were the sources of vehicle specifications. Figure 1 shows the BMW i4 structural make. Among the critical parameters are:

- Road-load coefficients: A = 31.9 lbf (141.9 N), $B = 0.03\ lbf/mph\ (0.299\ N \cdot sm^{-1})$, and $C = 0.01618\ lbf/mph^2\ (0.360\ N \cdot s^2/m^2)$
- EPA test weight: 4,750 $pounds$ (2,154.56 kg)
- Power rating: 282 $horsepower$ (210 kW)
- Motor torque: 400 Nm (295 $lb.ft$)
- Gear ratio: 1:1 (single-speed transmission)
- Usable battery capacity: 66 kWh.
- Powertrain efficiency: 86%
- Drag coefficient: 0.25
- Frontal area: 2.33 m^2

Figure 1. BMW i4 eDrive35 Gran Coupe

2. Grid Energy Consumption, Fuel Consumption, MPGe, and Regional WTW Emissions Simulation and Calculation

Three methodologies were employed to calculate vehicle energy consumption:

Method 1—Calculation Using Given Battery Capacity

In this method there is use of the rated battery capacity and the manufacturer's specified range, which it can go before being depleted, to compute the vehicle's grid energy consumption, fuel consumption, MPGe, and regional WTW emissions.

Grid energy available: $E_{grid} = E_{battery}/\eta_{powertrain}$

Adjusted combined fuel consumption: $FC_{adj} = E_{grid}/Range$

Adjusted combined MPGe:

$$MPGe = (33.705kWh/ga)/AdjCombined\ FC$$

Regional WTW emission:

$$gCO_{2,wtw} = regional\ emission\ factor * FC$$

Method 2—Simulation-Based Calculation

In this method there is use of a simulation model of the car, which runs over 1 cycle (1369 s) of the Urban Dynamometer Driving Schedule to get grid energy consumption, fuel consumption, MPGe, and regional WTW emissions. Figure 2 shows the simulation.

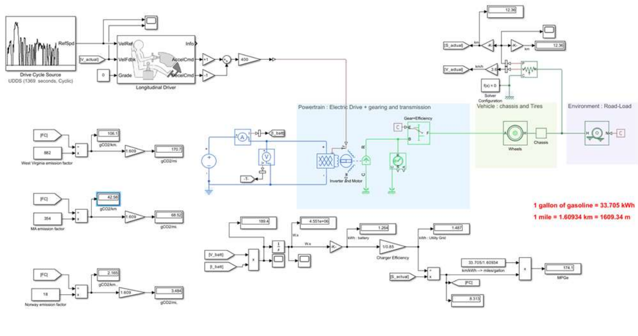

Figure 2. MPGE and Emission Factor Simulation

Method 3—Calculation Using Simulation Distance and Time

In this method the simulation time and the distance the model travels are noted and used to calculate grid energy consumption, fuel consumption, MPGe, and regional WTW emissions with the help of other parameters from the manufacturer. The assumption here is that the car is moving at a constant speed steadily on a flat road for the range and time.

Resistive Force: $F_{res} = A + Bv + Cv^2$

Power at wheels: $P = Fv = T\omega$

Rotational speed: $\omega = v/r$

Battery power: $P_b = P/(\eta_m * \eta_g * n_g)$

Battery energy used: $E_b = P_b * t$

Grid energy available: $Egrid = E_b/efficiency$

Fuel consumption: $FC = E_{grid}/Range$

MPGe: $MPGe = 0.7 * (33.705kWh/ga)/FC$

Regional WTW emission:

$$gCO_{2,wtw} = regional\ emission\ factor * FC$$

B. Phase 2: Inverter Design

1. System Specifications

The inverter was designed to meet the specific requirements:

- Topology: Three-phase, two-level voltage source inverter
- Output power: $210kW$
- Torque: $400Nm$
- DC bus voltage: $400V$ nominal (480V maximum, 750V transient)
- Modulation: SVPWM
- Switching frequency: 20 kHz
- Target efficiency: >98%

2. Operating Point Calculations

Motor mechanical speed: $\omega_{mech}= Power/Torque$

Electrical frequency (6-pole motor): $f_e = poles/2 * f_{mech}$

Maximum Phase Voltage: $V_{ph,max,(rms)} = V_{DC}/\sqrt{6}$

Peak Phase Voltage: $V_{ph,max,(Pk)} = \sqrt{2}\, V_{ph,max,(rms)}$

Maximum Phase Current:

$$I_{ph,\max(rms)} = \frac{P}{3V_{ph,\max(rms)}}$$

Peak Phase Current: $I_{ph,max,(Pk)} = I_{ph,max(rms)}\sqrt{2}$

3. Loss Calculations

Losses were evaluated at 100%, 50%, and 25% load using device-level RMS currents extracted from the simulated inverter leg waveforms. Because the Wolfspeed CAB650M17HM3 is an integrated SiC MOSFET module with an intrinsic anti-parallel diode, the total losses per device include MOSFET channel conduction, MOSFET switching energy, diode reverse-recovery loss, and residual dead-time conduction.

A. MOSFET Conduction Loss

The MOSFET conduction loss for both current directions is calculated using

$$P_{cond,MOS,total} = R_{DS(on)}\left[I^2{}_{MOS,rms2} + I^2{}_{D,rm}\ \ (1 - D_{dead})\right]$$

Using the RMS currents extracted from the full-load Simulink waveform and $R_{DS(on)} \approx 1.96\ m\Omega$, the module conduction loss is

$$P_{cond,MOS,total} = 174W$$

B. MOSFET Switching Loss

Switching losses are computed from datasheet energy values (or charge-based equivalents):

$$P_{sw} = \left(E_{on} + E_{off}\right)f_{sw}$$

At full load, the turn-on/turn-off switching losses sum to:

$$P_{sw} = 302W$$

This includes all voltage-current overlap and capacitive discharge contributions during commutation.

C. Body Diode Losses

Although SiC MOSFETs have very fast intrinsic diodes, there are two diode-related loss mechanisms:

1) Reverse-Recovery Loss

Reverse-recovery dissipation is significant at full load due to high di/dt:

$$P_{rr} = 75W$$

2) Dead-Time Conduction Loss

During dead time, the intrinsic diode briefly conducts.

$$P_{cond,body,diode} = V_f I_{D,avgDdead}$$

where V_f is the diode forward voltage and $I_{D,\text{avg}}$ is the average diode current when it conducts. For a typical full-load case, the idealized continuous-conduction diode loss is about 5.76 W; with a dead-time fraction of approximately

$$D_{dead} = 0.01$$

only about 1% of the cycle conducts through the diode, so\

$$P_{cond,diode} = 5.76W \times 0.01 = 0.06W$$

This term is negligible compared to other losses.

D. Total Device Losses and Inverter Efficiency

The combined semiconductor loss per SiC MOSFET module is:

$$P_{device} = P_{cond,MOS} + P_{sw} + P_{rr} + P_{dead} \approx 551\ W.$$

The three-phase inverter uses six active devices, giving:

$$P_{total} = 6 \times 551 = 3306W$$

For a 210 kW operating point:

$$\eta = \frac{P_{out}}{P_{out} + P_{total}} = \frac{210000}{210000 + 3306} = 98.45\%$$

This efficiency aligns with typical SiC automotive traction inverters operating at high load and high switching frequency.

4. Thermal Analysis

Junction temperatures were calculated using thermal resistance data. MOSFET and body diode junction temperature:

$$T_{J,Q} = T_{HS} + R_{JQ-HS} * P_{MOS}$$
$$T_{J,D} = T_{HS} + R_{JD-H} * P_D$$

5. System Simulation

The inverter topology was simulated to improve reliability and help in analysis. Figure 3 shows the inverter circuit, control, and thermal dissipation.

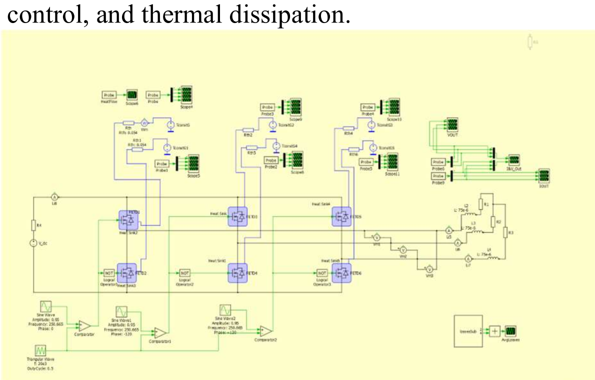

Figure 3. Inverter Simulation

6. Component Selection

The active switch chosen was the Wolfspeed CAB650M17HM3 SiC MOSFET module, with a blocking voltage of 1700 V and continuous current of 650 A. On-resistance ($R_{DS(on)\,@\,25°C}$) of 1.82 mΩ (full load), 1.45 mΩ (half load), 1.2 mΩ (quarter load) and gate charge of 375 nC, This device was selected based on voltage margin (1700 V rating vs. 750 V maximum), current capability (exceeding maximum of 606A), low on-resistance, and integrated anti-parallel diode. Figure 4 shows the MOSFET physical structure and its circuit configuration.

Figure 4. Wolfspeed CAB650M17HM3 SiC MOSFET

C. Phase 3: Battery Charger Design

1. Charger Requirements

The BEV selected for this project is the BMW i4, which uses BMW's 5th generation eDrive electric platform. The vehicle is equipped with a nominal 400V lithium-ion battery pack, with a pack voltage typically ranging from approximately 380V to 430V depending on state of charge.

For this project, an on-board AC charger is designed to support single-phase level 2 charging. In North America, level 2 charging is commonly supplied by 240V AC, and the BMW i4 supports AC charging to 7.3kW.

The resulting design requirements are:

- Input Voltage: 240V AC (single-phase, 60Hz)
- Target Output Voltage: 400V DC nominal.
- Maximum Output Power: 7.3kW
- Required Output Current:

$$I_{out} = \frac{P_{out}}{V_{out}} = \frac{7300}{400} = 18.25A$$

- Power Factor Requirement: PF ≈ 1 using a PFC front-end
- Isolation Requirement: Mandatory DC-DC galvanic isolation via a high-frequency transformer

These specifications define the electrical operating conditions for the charger design in Phase 3 and serve as the basis for selecting the charger topology, components, and control strategy.

2. Charger Topology

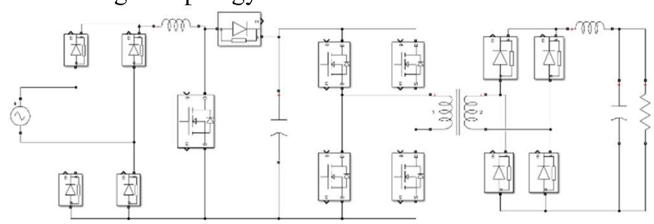

Figure 5. Overall Design of the System

The on-board charger designed for the BMW i4 follows a two-stage isolated AC–DC architecture, which is widely adopted in commercial EV chargers. The model implemented in Simulink consists of:

- A Boost Power Factor Correction (PFC) front-end, which ensures unity power factor and regulates the DC-link voltage.
- A full-bridge PWM isolated DC–DC converter, which steps the PFC output to the required 320–430 V battery charging range.

AC–DC Stage: Boost PFC

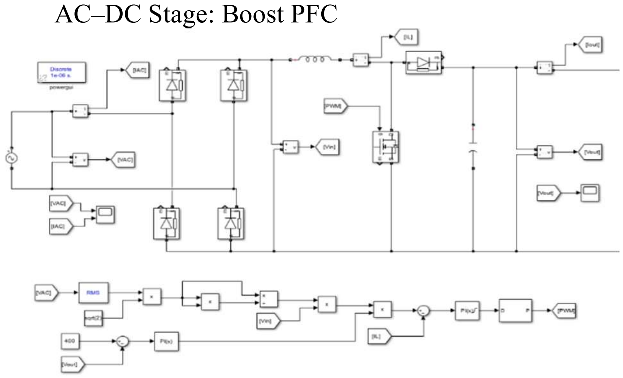

Figure 6. Boost PFC Design.

The first stage is a Boost PFC rectifier, consisting of:

- Full-bridge diode rectification
- Boost inductor
- Active switching device (IGBT/MOSFET)
- DC-link output capacitor
- Current and voltage sensing blocks
- A cascaded PI voltage loop and current loop controller

The operating principle of this part is that the AC input is rectified into an unregulated DC waveform. The boost switch operates at high frequency to shape the inductor into a sinusoidal waveform, maintain a high-power factor, and regulate the DC link voltage at 380-400V.

The main reason why we choose Boost PFC is that it has a simple and robust topology. It can be easily controlled with average current mode control and is widely used in EV chargers around 7.3kW. Since it has lower complexity compared to Totem-Pole PFC, it is more suitable for households. This stage delivers a stable DC signal which feeds the isolated converter.

DC–DC Stage: Full-Bridge PWM Converter

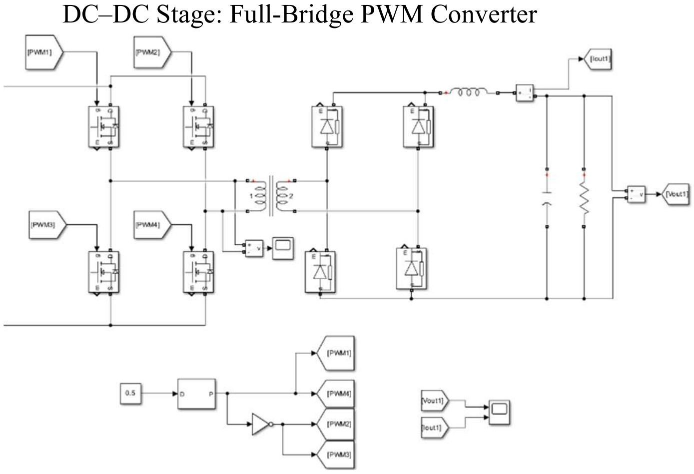

Figure 7. Full-Bridge Converter Design.

The second stage is a hard-switched full-bridge PWM isolated converter, consisting of:

- Four primary-side MOSFETs forming an H-bridge
- High-frequency transformers (providing galvanic isolation)
- Secondary-side full-bridge synchronous rectifier
- Output LC filter
- Voltage and current measurement for feedback

Because the battery voltage varies throughout the charging cycle, the converter regulates its output using open-loop PWM control. The measured output voltage is compared with a 400-V reference, and a PI controller modulates the duty ratio of the full bridge to maintain the desired charging voltage and current. The complete charger therefore operates as a coordinated system in which the PFC stage ensures unity-power-factor AC input and a stable DC-link voltage, while the isolated DC–DC converter provides the necessary voltage regulation, isolation, and power processing to charge the vehicle battery. The overall topology is well suited to the 7.3 kW charging requirement of the BMW i4 and aligns closely with the structures used in commercial on-board chargers. Figure 5 shows the full Simulink implementation of this topology, including the power stages, transformer, synchronous rectification, output filter, and dual-loop control structure.

3. Component Selection

The key power semiconductor devices selected for the on-board charger are the STTH6010-Y ultrafast diode and the Infineon IPDQ65R008CM8 650V CoolMOS™ CM8 MOSFET. Both devices were chosen to match the demanding requirements of a 7.3kW isolated full-bridge converter operating from a 400V DC link and targeting high efficiency, low switching loss, and robust thermal performance.

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a. MOSFET Selection – Infineon IPDQ65R008CM8

The primary-side full-bridge employs the IPDQ65R008CM8, a state-of-the-art 650V superjunction MOSFET with an exceptionally low on-state resistance of 8mΩ at 25°C. The extremely low $R_{ds(on)}$ directly reduces conduction loss during high current operations. This is particularly important because the full-bridge must process the entire 7.3kW output, with primary-side peak currents exceeding 60-80 A during switching transitions. Aside from conduction loss, the MOSFET offers favorable dynamic behavior. Its total gate charge is 375 nC, and the turn-on and turn-off times are 49 ns and 218 ns, respectively [8] (page 6), enabling efficient high frequency switching without excessive switching loss. The CoolMOS™ CM8 platform also integrates a fast body diode, capable of reverse recovery times around 280–350 ns, with a peak recovery current of 18 A [8] (Table 7). This fast body diode is particularly beneficial in full-bridge operation, where commutation between MOSFET legs frequently forces the intrinsic diode to conduct briefly.

Thermally, the device is suited for high-power applications. Its junction-to-case thermal resistance is 0.1 K/W [8] (Table 3), allowing effective heat extraction even under several tens of watts of loss.

Overall, the MOSFET was chosen because it balances low conduction loss, manageable gate charge, strong avalanche and dv/dt ruggedness, and superior thermal characteristics, making it an ideal choice for both the PFC stage and the high-frequency full-bridge converter.

b. Diode Selection: STTH6010-Y Ultrafast Recovery Diode

STTH6010-Y was selected as the main high-power ultrafast diode due to its automotive qualification and excellent dynamic characteristics. The diode supports a 60A forward current and an RMS current of 80A, with a repetitive peak forward capability of 450A [9] (Table 2). Such ratings provide a substantial margin under high-current charging and peak transient events during mode transitions. The typical forward voltage drop of 1.3-1.4V at rated current [9](page 2) help limit conduction losses, while the device's reverse recovery time ranges from 49-61 ns under typical operating conditions [9](Table 5). This fast recovery significantly reduces switching losses and mitigates voltage overshoot on the secondary side.

STTH6010-Y's thermal resistance of 0.78°C/W [9](Table 3) ensures adequate heat dissipation even in compact packaging. Coupled with its high thermal robustness (max junction temperature 175°C), the device is well suited to operate in the elevated temperature environment common inside automotive OBC housings.

Together, the selected MOSFETs and Diodes provide:

- Voltage headroom above the 400V nominal DC bus (MOSFET: 650V, diode: 1000V)
- Current capability is consistent with high-power full-bridge and secondary rectification demands.
- Low switching and conduction loss that help meet the overall charger efficiency target
- Automotive-grade reliability

These characteristics make the component set well matched to the electrical, thermal, and environmental requirements of the BMW i4's 400V charging platform. The combination of a low-loss superjunction MOSFET and a high-speed, high-current diode provides a robust foundation for efficient AC/DC conversion across the full operating range.

4. Passive Component Design

The design of the 7.3kW charger was carried out by determining the key passive elements and transformer parameters needed for stable operation of the Boost PFC and the full-bridge isolated DC-DC stage. For the PFC stage, the boost inductor was sized based on the allowable inductor current ripple in continuous-conduction mode. Using the standard approximation

$$L = \frac{\sqrt{2V_{ph}}}{f_s \Delta I_{L,p-p}}$$

where fs is the switching frequency and Δi_L is the desired ripple, an inductance of approximately around 300 µH provides stable operation at 7.3 kW with a 240-V input. For the isolated DC-DC stage, the transformer turns ratio was selected to align the nominal operating duty cycle with efficient full-bridge switching. The ratio is chosen around 1:1, ensuring that the converter operates with a duty cycle close to 0.5 when producing 400 V at the output. These modeled component values were directly implemented in Simulink, forming the basis of the closed-loop control design and enabling realistic loss calculations using the datasheet parameters introduced in the Component Selection.

IV. RESULTS

A. Phase 1: Vehicle Performance and Emissions

1. Energy Consumption and Efficiency

Table 1 summarizes energy consumption results from the three methodologies:

Metric	Battery Capacity Method	Simulation (UDDS)	Simulation time and distance
$E_{grid}\ (kWh)$	76.74	1.49	1.33
$Power\ (kW)$	-	3.37	3.5
FC (kWh/100mi)	25.20	8.31	9.48
MPGe	134.20	174.10	198.5

Table 1. Energy Consumption Comparison

The manufacturer's stated MPGe of 155.1 falls between the conservative battery capacity method (134.2 MPGe) and the UDDS simulation (174.1 MPGe). The EPA-certified value of 167 MPGe represents a weighted combination of city and highway cycles with adjustment factors.

The simulation-based consumption (8.31kWh/100mi over the UDDS cycle) represents gentle urban driving at lower speeds where aerodynamic losses are minimal. This demonstrates strong dependency of BEV efficiency on driving patterns.

2. Regional Emissions Analysis

Table 2 summarizes regional emissions analysis results from the three methodologies:

Region	Battery Capacity Method	Simulation (UDDS)	Simulation time and distance
Massachusetts (354gCO2/kWh)	89.20	68.52	60.78
West Virginia (882gCO2/kWh)	22.26	170.7	149.70
Norway (20gCO2/kWh)	5.04	3.48	3.06

Table 2. Regional Emissions Analysis Comparison

Key findings:

Emissions in coal-intensive West Virginia are 44 times higher than renewable energy-powered Norway, demonstrating that BEV environmental benefits depend critically on the electricity source.

The BMW i4 in Massachusetts produces 70-71% lower emissions, West Virginia 26% lower emissions, and in Norway 98% lower emissions compared to a typical gasoline vehicle.

The battery capacity method yields higher estimates as it assumes average efficiency across all driving conditions. Simulation over specific cycle shows 20-25% lower emissions due to the favorable characteristics of UDDS cycle.

B. Phase 2: Inverter Performance

The performance was found by calculating and simulating as in the methodology. The following plots show the simulation results for the three different loads.

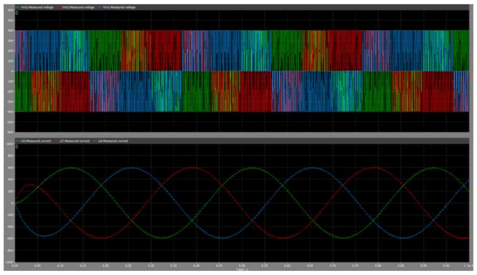


Figure 8. Output Phase Voltage and Current at 100% Load

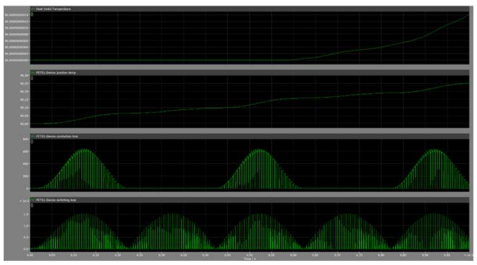


Figure 9. Junction and Heat Sink Temperatures, Switching and Conduction Losses at 100% Load.

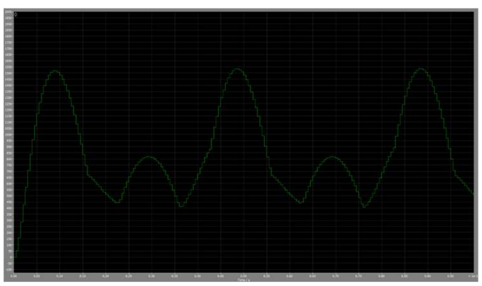


Figure 10. Average Total Losses at 100% Load.

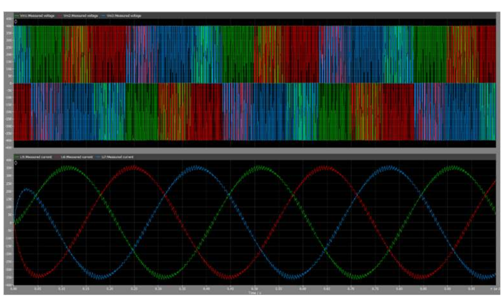


Figure 11. Output Phase Voltage and Current at 50% Load

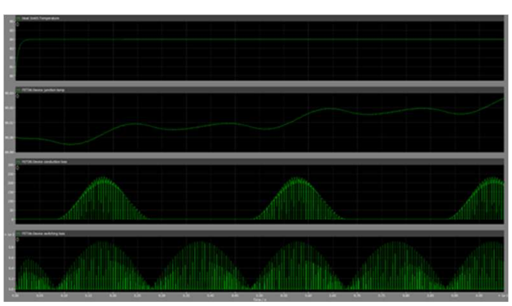


Figure 12. Junction and Heat Sink Temperatures, Switching and Conduction Losses at 50% Load

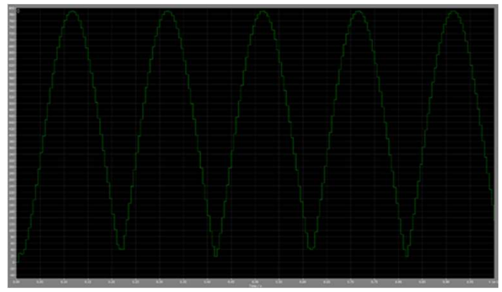


Figure 13. Average Total Losses at 50% Load

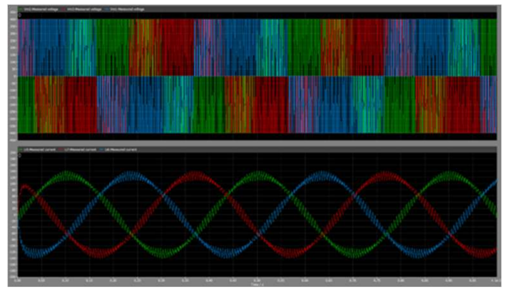


Figure 14. Output Phase Voltage and Current at 25% Load

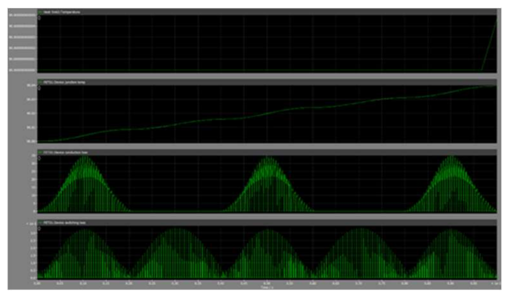


Figure 15. Junction and Heat Sink Temperatures, Switching and Conduction Losses at 25% Load

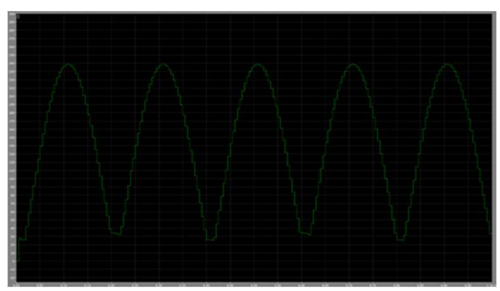


Figure 16. Average Total Losses at 25% Load.

Table 3 summarizes the inverter performance at three different loads using calculations, while table 4 summarizes the inverter performance at three different loads using simulation.

Parameter	100% Load	50% Load	25% Load
Output Power (kW)	210	105	52.5
Max Phase Current RMS (A)	428.66	214.33	107.17

Max Phase Voltage RMS (V)	163.299	163.299	163.299
Power Factor	0.95	0.93	0.91
Modulation Index	0.95	0.95	0.95
MOSFET losses (W)	477	194	91
Diode Losses (W)	65.31	37.03	18.15
Total Losses (W)	3306	1386	654
Efficiency	98.45%	98.70%	98.77%
$T_{j,D}$	94.00	92.00	90.98
$T_{j,Q}$	115.8	100.5	95

Table 3. Inverter Electrical Performance Using Calculations

Motors have a characteristic of producing more reactive power at reduced loads which led to the assumption of lower power factors at lower loads. The modulation index chosen was 0.95 to leave 5% margin below the SVPWM limit to avoid over-modulation.

Parameter	100% Load	50% Load	25% Load
Output Power (kW)	210	105	52.5
Phase Current Peak (A)	600	350	140
Phase Voltage Peak (V)	400	400	400
Power Factor	<1	<1	<1
Modulation Index	0.95	0.95	0.95
Total Losses (W)	4650	2340	705
Efficiency	97.833%	97.820%	98.675%
Heat sink	120	100	92
$T_{j,avg}$	110	110	96

Table 4. Inverter Electrical Performance Using Simulation

The simulation for the SiC MOSFET was for the integrated module (MOSFET and body diode) therefore, MOSFET losses and diode losses could not be separated.

At full loads, switching and conduction losses are nearly equal for MOSFETs. At reduced loads, switching losses dominate because they scale with voltage and frequency but not directly with current.

Efficiency improves at partial load, reaching 98.77% at 25% load, and lowers as load increases. These favorable characteristic benefits real-world driving, where average power is well below peak.

Body diode losses are relatively small due to minimal conduction time in SVPWM. Most current flows through the MOSFET channel due to synchronous rectification.

Heat sink temperature was assumed at 90°C, representing realistic liquid cooling performance. Maximum junction temperatures remain well below the 175°C rating of SiC devices, providing adequate thermal margin. The 25.8°C temperature rise in MOSFETs at full load confirms adequate heat sinking.

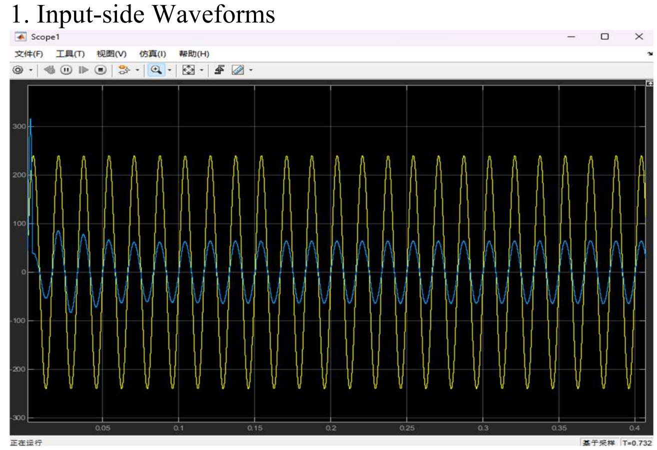
The simulation shows higher losses than analytical calculations. This discrepancy arises from:

- More conservative component models in simulation.
- Inclusion of additional parasitic losses (gate drive, capacitive switching).
- Dynamic effects during transients are not captured in steady-state analysis.

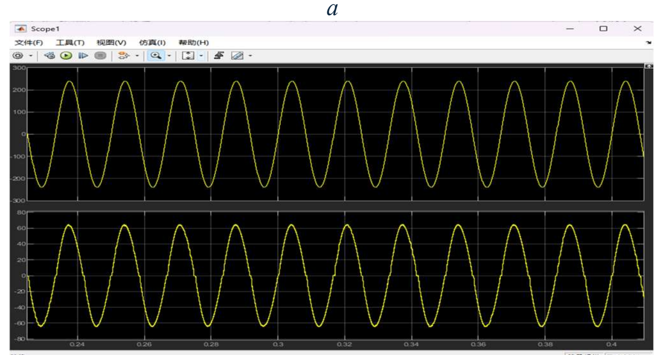
Both methods confirm efficiency above 97%, validating the design approach.

C. Phase 3: Battery Charger Performance

1. Input-side Waveforms



a



b

Figure 17. a: V_{ac}, I_{ac} Waveform; b: Zoomed V_{ac}, I_{ac} Waveform.

At the input of the charger, the simulated AC voltage and current waveforms exhibit the expected sinusoidal behavior. The input current closely follows the shape of the 240V AC line voltage and is nearly in phase with it, demonstrating correct operation of the Boost PFC stage. The absence of noticeable distortion and the strong phase alignment confirm that the average current mode controller successfully forces the input current to track the AC voltage, resulting in near–unity power factor performance. These waveforms verify that the PFC stage is properly shaping the input current and supplying a stable, low-ripple DC link voltage to the downstream full-bridge converter.

2. DC-link Voltage

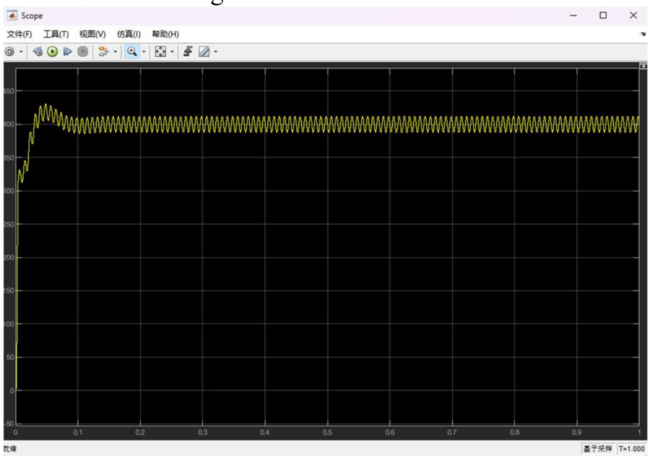


Figure 18. DC link voltage.

The DC-link voltage stabilizes around 395-400 V with only minimal peak-to-peak variation. This low-ripple behavior indicates that the bulk capacitor provides adequate energy buffering during the rectification and boosting process. Together, these waveforms verify that the PFC stage maintains stable intermediate-bus conditions and supplies clean DC power to the downstream full-bridge converter under steady-state operation.

3. Output Voltage and Current Regulation

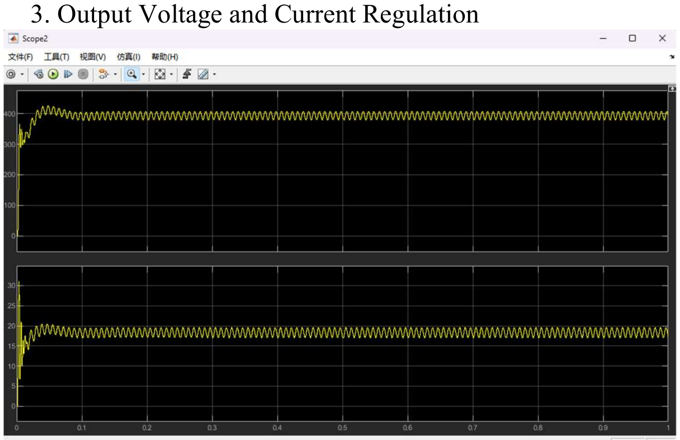


Figure 19. Output Voltage and Current Waveform.

The closed-loop control structure of the charger demonstrates stable output regulation during full-power operation. The voltage controller maintains the output near the 400V reference with negligible steady-state error, and the inner current loop responds rapidly to changes in load and switching conditions. During startup, the output voltage rises smoothly without significant overshoot, indicating proper tuning of both the PI voltage regulator and the current-mode control in the DC–DC stage. Under steady-state conditions, the output current stabilizes at approximately 18-19 A, corresponding to the rated 7.3kW charging level for the BMW i4 battery system. The combination of well-damped transient behavior, low ripple, and accurate reference tracking confirms that the dual-loop control architecture is effective in ensuring consistent charging performance across the operating range. These results verify that the system meets the functional requirements of the designed on-board charger and provides stable and reliable output suitable for automotive battery charging.

4. Loss Analysis

A comprehensive loss analysis was performed for the 7.3kW on-board charger by separating the power dissipation into the Boost PFC stage and the isolated full-bridge DC-DC converter. The objective of this evaluation is to determine the conduction losses of the selected MOSFETs and diodes, estimate switching losses based on device capacitances and reverse-recovery charge, and validate the results using Simulink-based semiconductor models incorporating datasheet parameters. Although the MOSFET datasheet does not provide explicit turn-on and turn-off energy (Eon/Eoff) values, the dynamic losses can be accurately approximated using output capacitance (Coss) and charge-based models, which is the standard method for automotive-grade superjunction MOSFETs.

PFC Stage Losses

The conduction loss per rectifier diode is

$$P_{R,cond} = 4 * [V_{f0} I_{R,dc} + r_f I_{R,rms}^2]$$

The MOSFET conduction loss is

$$P_{Q,cond} = R_{DS,on} I_{Q,rms}^2$$

The turn-on and turn-off power loss in the switch is

$$P_{Q,sw} = f_s (E_{on} + E_{off}) \frac{V_{dc}}{V_{test}}$$

The diode conduction loss is

$$P_{D,cond} = V_{f0} I_{D,dc} + r_f I_{D,rms}^2$$

The inductor loss is

$$P_L = R_{cu} I_{ph}^2$$

After combining conduction and switching components, the PFC stage loss is dominated by MOSFET conduction and diode forward loss, with capacitive switching loss representing a smaller but non-negligible fraction. This aligns with typical PFC performance at high power levels.

Isolated Full-Bridge DC–DC Converter Side

The MOSFET conduction loss is

$$P_{Q,cond} = R_{DS,on} I_{Q,rms}^2$$

Combining all conduction, capacitive switching, and reverse-recovery losses yields an overall semiconductor loss profile consistent with typical automotive chargers of similar topology. The PFC stage is dominated by conduction loss in the MOSFET and boost diode, while the DC-DC stage experiences a mix of conduction and switching losses across the full-bridge switches and rectifiers. Based on the estimated loss values observed in Simulink, the efficiency of the boost PFC converter falls to 97.37%, and the efficiency of the DC-DC converter falls to 95.32%, which results in a total system efficiency of 92.82% at the rated 7.3 kW output, which is consistent with practical 400 V onboard chargers used in production vehicles. These results validate both the component selection and the electrical design presented in earlier sections.

Output Voltage (V)	Target Power (kW)	Output Power (kW)	PFC Conduction Loss (W)	PFC Switching Loss (W)	PFC Efficiency	DC-DC Conduction Loss (W)	DC-DC Switching Loss (W)	DC-DC Efficiency	Overall Efficiency
393.8	7.3	7.219	108.266	169.774	97.37%	81.1	97.9	95.32%	92.82%

Table 5. Losses and Efficiency

The overall performance of the designed 7.3 kW on-board charger meets the functional and electrical requirements established in Phase 3. The Boost PFC stage achieves near-unity power factor operation while maintaining a stable 400V DC link voltage under full-load conditions. The isolated full-bridge DC-DC converter produces a regulated 400V output with an average charging current of approximately 18-19 A, matching the rated 7.3 kW output power of the BMW i4 charging platform.

The loss analysis confirms that the semiconductor dissipation in both the PFC and DC-DC stages remains within acceptable limits, resulting in an overall estimated efficiency of 92.82%, which is consistent with typical automotive on-board chargers of similar architecture. The combined simulation and analysis results validate the correctness of the topology choice, component selection, and control strategy implemented in this design.

V. CONCLUSION

This report presented a comprehensive modeling and simulation study of the powertrain and power electronic subsystems for the 2025 BMW i4 eDrive35 Gran Coupe. By integrating vehicle dynamics with detailed electrical designs, the study validated the performance of the traction inverter and on-board charger under realistic operating conditions.

First, the vehicle characterization analysis highlighted the critical relationship between grid composition and environmental impact. While the simulation under the UDDS drive cycle demonstrated a high vehicle efficiency of **174.1 MPGe**, the well-to-wheel analysis revealed significant regional disparities. Emissions in coal-intensive regions were found to be approximately 44 times higher than in renewable-dominated regions, underscoring the necessity of grid decarbonization to realize the full environmental potential of BEVs.

Second, the design of the 210 kW three-phase inverter confirmed the superior performance of Silicon Carbide (SiC) technology. Utilizing Wolfspeed SiC MOSFETs and Space Vector PWM (SVPWM), the design analysis demonstrated a peak efficiency of **98.45** at full load, with simulation results confirming high-efficiency operation across the operating range. Thermal analysis further verified the robustness of the system, with average junction temperatures stabilized at **110°C** under full power, well below the device's thermal limits.

Finally, the 7.3 kW On-Board Charger (OBC) was successfully modeled using a two-stage topology consisting of a Boost PFC and an isolated Full-Bridge DC-DC converter. The system achieved a near-unity power factor and a stable 400V DC output with minimal ripple. The comprehensive loss analysis yielded an overall charger efficiency of **92.82%**, meeting the functional and efficiency requirements for Level 2 automotive charging systems.

In conclusion, the results validate that the proposed designs meet the stringent electrical and thermal requirements of the BMW i4 platform. The successful integration of wide-bandgap devices and efficient control strategies provides a solid foundation for future hardware implementation.

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APPENDIX

Appendix A: Detailed Calculations

A. Utilizing given Battery capacity

$$E_{grid} = \frac{Ebattery}{Powertrain\ Efficiency} = \frac{66k}{0.86}$$

= 76.74 kWh

$$Combined\ \ Adj\ MPGe = \frac{33.705}{Adj\ Combined\ FC} = \frac{33.705}{\frac{25.2}{100}}$$

$$= 134.2 MPGe$$

Adj Combined FC = $\frac{Egrid}{range} = \frac{76.74}{304.5}$ = 0.252kWh/mi =25.2kWh/100mi

$$In\ MA, gCO2_{wtw} = 354 gCO_2/kWh * 25.2 kWh/100mi$$
$$= 89.20 gCO2/mi$$

$$In\ WV, gCO2_{wtw} = 882 gCO_2/kWh * 25.2 kWh/100mi$$

$$= 222.26 gCO2/mi$$

$$In\ Norway, gCO2\_wtw = 20 gCO2/mi * 25.2 kWh/100mi$$

$$= 5.04 gCO2/mi$$

B. Utilizing simulation Range and Time

$$t_{drivecycle} = 1369s = 0.38h;$$

$$Range_{drivecycle} = 12.33km = 7.66mi$$

$$Vavg = \frac{7.66mi}{0.38028h} = 20.15mph = 9m/s$$

$$Fres = A + Bv + Cv^2$$

$$= 131.4N + 0.299N/ms - 1 * 9m/s + 0.360N/m2s - 2 * 92m2/s2 = 163.3N$$

$$P = Fv = T\omega = 163.319 * 9 = 1.471kW$$

$$\omega = v/r = (9m/s)/0.292m = 30.856rad/s$$
$$= 294.655rpm; T = P/\omega$$
$$= 47.636Nm$$

$$Pb = P/(\eta m * \eta g * ng)$$

$$= 1471W/(0.9 * 0.96 * 1) = 1.702kW$$

$$Eb = Pb * t = 1.702kW * 0.38h = 0.6470kWh$$

$$Egrid = 0.6470/0.86 = 0.7523kWh$$

$$FC = 0.7523kWh/7.66mi$$

$$= 0.09821kWh/mi = 9.821kWh/100mi$$

$$MPGe = 0.7 * (33.705kWh/ga)/(0.09821kWh/mi)$$

$$= 0.7 * 343.2\ mpg = 240.22\ mpg$$

$$In\ MA, gCO2\_wtw = 354 gCO2/kWh * 9.82kWh/100mi$$

$$= 34.8 gCO2/mi$$

$$In\ WV, gCO2\_wtw = 882 gCO2/mi * 9.82kWh/100mi$$

$$= 86.62 gCO2/mi$$

$$In\ Norway, gCO2\_wtw = 20 gCO2/mi * 9.82kWh/100mi$$

= 1.96gCO2/mi

C. Inverter Calculations

$$ma = \frac{2\sqrt{2}\,V_{ph}}{V_{dc}}; Assume\ ma = 0.95$$

$$D_{an} = \frac{1}{2} + \frac{m}{2}\cos(wt) = \frac{1}{2} + \sqrt{2}\frac{V_{ph}}{V_{dc}}\cos(wt)$$

$$= \frac{1}{2} + 0.57735\cos(wt)$$

$$D_{bn} = \frac{1}{2} + \sqrt{2}\frac{V_{ph}}{V_{dc}}\cos\left(wt - \frac{2\pi}{3}\right)$$

$$D_{cn} = \frac{1}{2} + \sqrt{2}\frac{V_{ph}}{V_{dc}}\cos\left(wt - \frac{4\pi}{3}\right)$$

$$w_{mech} = 2\pi f_{mech}\frac{P}{\tau} = \frac{210*10^3 W}{400NM} = 525 rad\ s^{-1}$$

$$f_{mech} = \frac{525}{2\pi}Hz = \frac{525*60}{2\pi}RPM = 5013.38RPM$$

$$= 83.556Hz$$

$$f_e = p * f_{mech} = \frac{6}{2}*83.556 = 250.668\text{Hz}$$

$$V_{ph,\max(rms)} = \frac{V_{dc}}{\sqrt{6}} = \frac{400}{\sqrt{6}} = 163.30V$$

$$V_{ph,\max(pk)} = 163.30\sqrt{2} = 230.94V$$

At 100% Load.

$$I_{ph,\max(rms)} = \frac{P}{3V_{ph,\max(rms)}} = \frac{210*10^3}{3*163.3} = 428.66\text{A}$$

$$I_{ph,\max(pk)} = 428.66\sqrt{2} = 606.215A$$

$$I_{Q(dc)} = I_{ph}\left(\frac{1}{\sqrt{2\pi}} + \frac{V_{ph}}{2V_{dc}}\cos(\varphi)\right) = 428.66 * \left(\frac{1}{\sqrt{2\pi}} + \frac{163.3}{2*400} * 0.95\right) = 179.608\text{A}$$

$$I_{D(dc)} = I_{ph}\left(\frac{1}{\sqrt{2}\pi} - \frac{V_{ph}}{2V_{dc}}\cos(\varphi)\right) = 428.66*\left(\frac{1}{\sqrt{2}\pi} - \frac{163.3}{2*400} * 0.95\right) = 13.357\text{A}$$

$$I_{HV(dc)} = 3\left(\frac{V_{ph}}{V_{dc}} I_{ph}\cos(\varphi)\right) = 3 * \left(\frac{116.6}{400} * 428.66 * 0.95\right) = 498.751\text{A}$$

$$I_{Q(rms)} = I_{ph}\sqrt{\frac{1}{4} + \frac{4\sqrt{2}V_{ph}}{3\pi V_{dc}}\cos(\varphi)}$$

$$= 28.66\sqrt{\frac{1}{4} + \frac{4\sqrt{2}*163.3}{3\pi*400} * 0.95} = 297.844\text{A}$$

$$I_{D(rms)} = I_{ph}\sqrt{\frac{1}{4} - \frac{4\sqrt{2}V_{ph}}{3\pi V_{dc}}\cos(\varphi)}$$

$$= 428.66\sqrt{\frac{1}{4} - \frac{4\sqrt{2}*163.3}{3\pi*400} * 0.95} = 56.244A$$

$$I_{HVdc(rms)} = I_{ph}\sqrt{\frac{\sqrt{6}}{\pi} * \frac{V_{ph}}{V_{dc}}(1 + 4\,cos^2\,(\varphi))}$$

$$=428.66\sqrt{\frac{\sqrt{6}*163.3}{\pi*400} * (1 + 4 * 0.95^2)} = 519.265\text{A}$$

$$I_{CHV(rms)} = I_{ph}\sqrt{\frac{\sqrt{6}*V_{ph}}{\pi*V_{dc}} + \left(\frac{4\sqrt{6}*V_{ph}}{\pi*V_{dc}} - 9\left(\frac{V_{ph}}{V_{dc}}\right)^2\right)\cos^2\varphi}$$

$$= 428.66\sqrt{\frac{\sqrt{6}*163.3}{400\pi} + \left(\frac{4\sqrt{6}*163.3}{400\pi} - 9\left(\frac{163.3}{400}\right)^2\right)0.95^2}$$

$$=144.512\text{A}$$

At 50% load and 25% load the same calculations are done with corresponding values.

Appendix B: Component Datasheets Summary

B.1 Wolfspeed CAB650M17HM3

Absolute Maximum Ratings:

- Drain-Source Voltage: 1700 V
- Continuous Drain Current ($T_C = 25°C$): 650 A
- Pulsed Drain Current: 1950 A
- Maximum Junction Temperature: 175°C

Electrical Characteristics ($T_J = 25°C$):

- $R_{DS(on)@\,V-GS} = 15V, I_D = 325A: 1.82\ m\Omega\ (typ), 2.5\ m\Omega\ (max)$

Gate-Source Threshold Voltage: 2.0-3.5 V

- Total Gate Charge: 375 nC
- Output Capacitance: 1200 pF
- Body Diode Forward Voltage: 1.3V at 325A

Thermal Characteristics:

- Junction-to-Case Thermal Resistance: 0.054°C/W

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